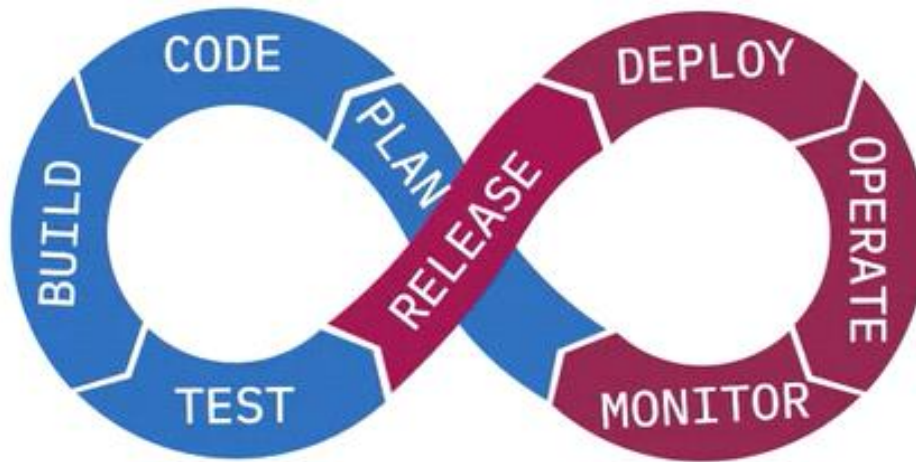


What is CI/CD

Wednesday, November 19, 2025

11:31 AM



1. CI/CD is not one process – it is two distinct processes
2. CI is continuously integrating code back to main/master branch
3. CD is deploying that integrated code somewhere

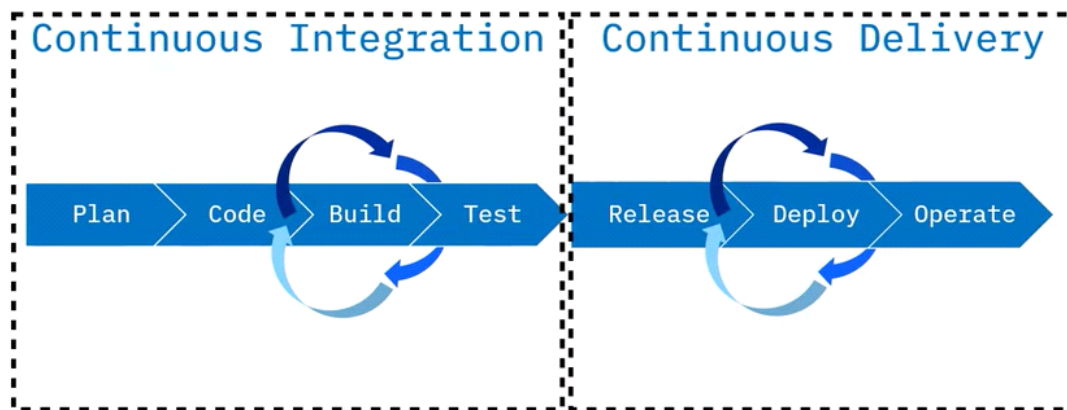
Continuous Integration:

1. Automation process that allows developers to **integrate** their work into a **repository**
2. Helps to **enable collaborative development** across teams
3. Helps to **identify integration bugs** sooner
4. Teams can work in **small chunks** and regularly **integrate** new code into **main branch**

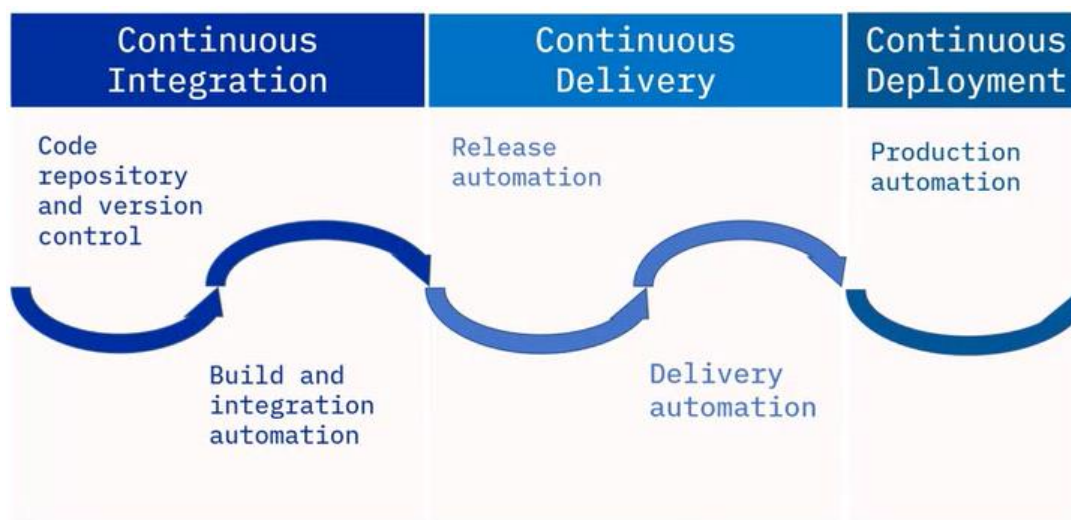
Continuous Delivery:

1. Continuous delivery comes after continuous integration
2. **Prepares** the **code** for **release**
3. **Automates** the **steps** that are needed to **deploy a build**

CI/CD Phases:



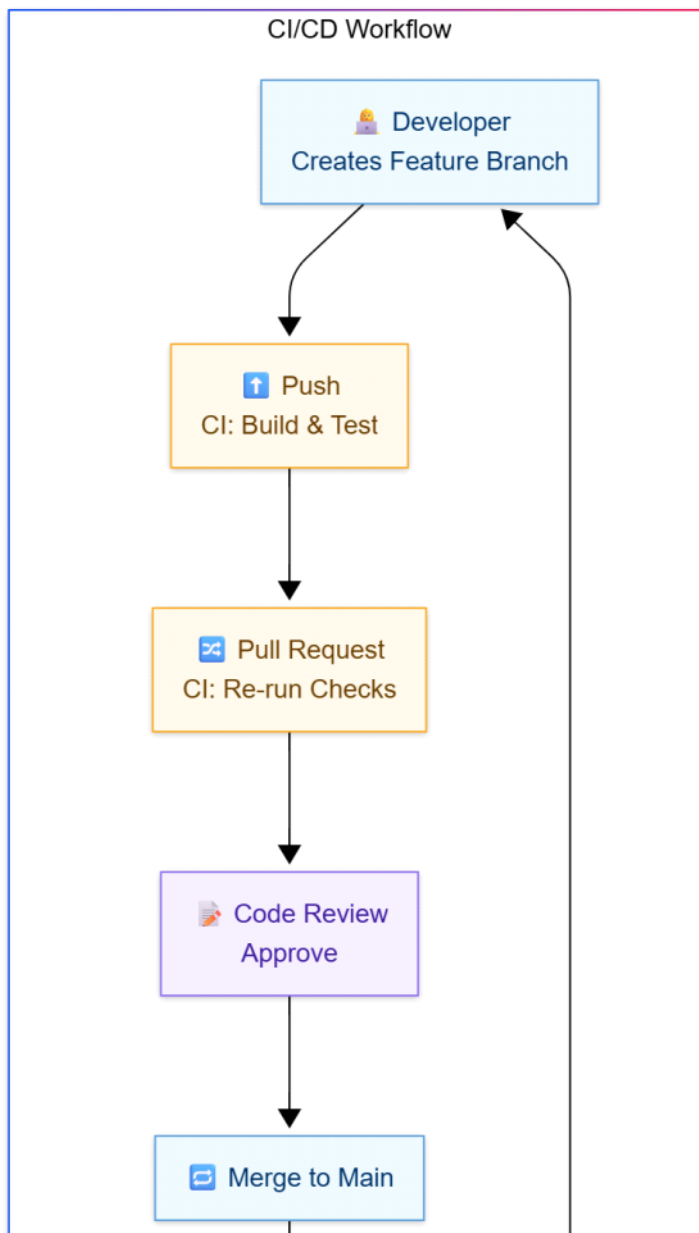
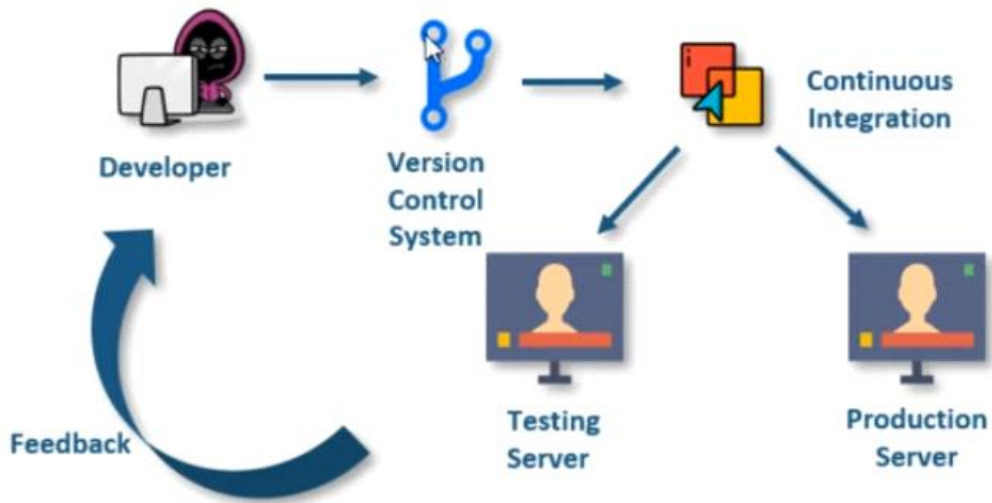
Continuous Deployment:

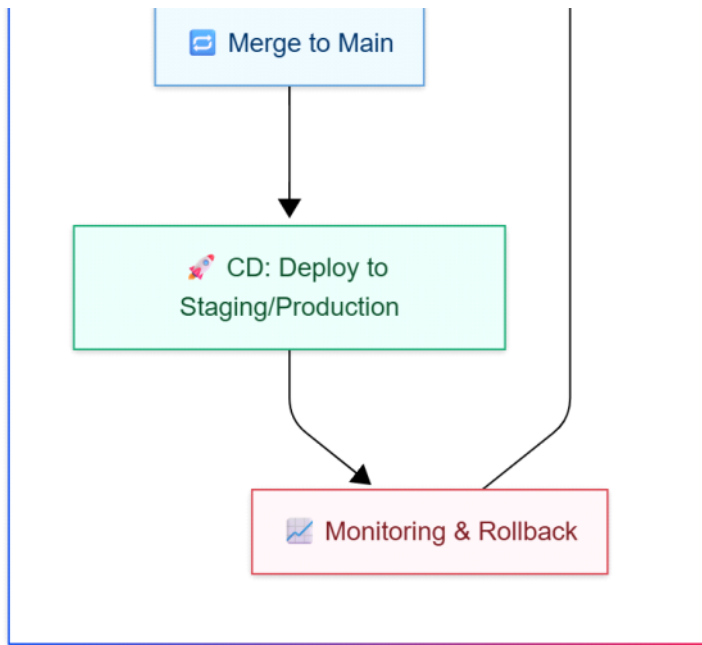


Benefits of CI/CD:

1. Faster reaction times to changes
2. **Higher code quality**
3. Code in version control works
 - a. **Code** in master/main branch is always **deployable**
4. **Less deployment time**
5. Reduced code integration risk
 - a. Integrating smaller changes
 - b. No need to integrate thousands of line of codes into a code base
 - c. Only need to integrate 10 – 50 lines of code

Flow of CI/CD





Platform and Tools

Wednesday, November 19, 2025 11:34 AM

Self-Hosted:

1. Jenkins
2. Bamboo
3. Teamcity
4. GoCD
5. Github Actions

Software as a Service (SaaS):

1. Travis CI
2. Codeship
3. Circle CI
4. Github Actions

Cloud Service Providers:

1. AWS: CodePipeline & CodeBuild
2. Azure Pipeline
3. GCP cloud Pipeline (Build)

GitHub Actions

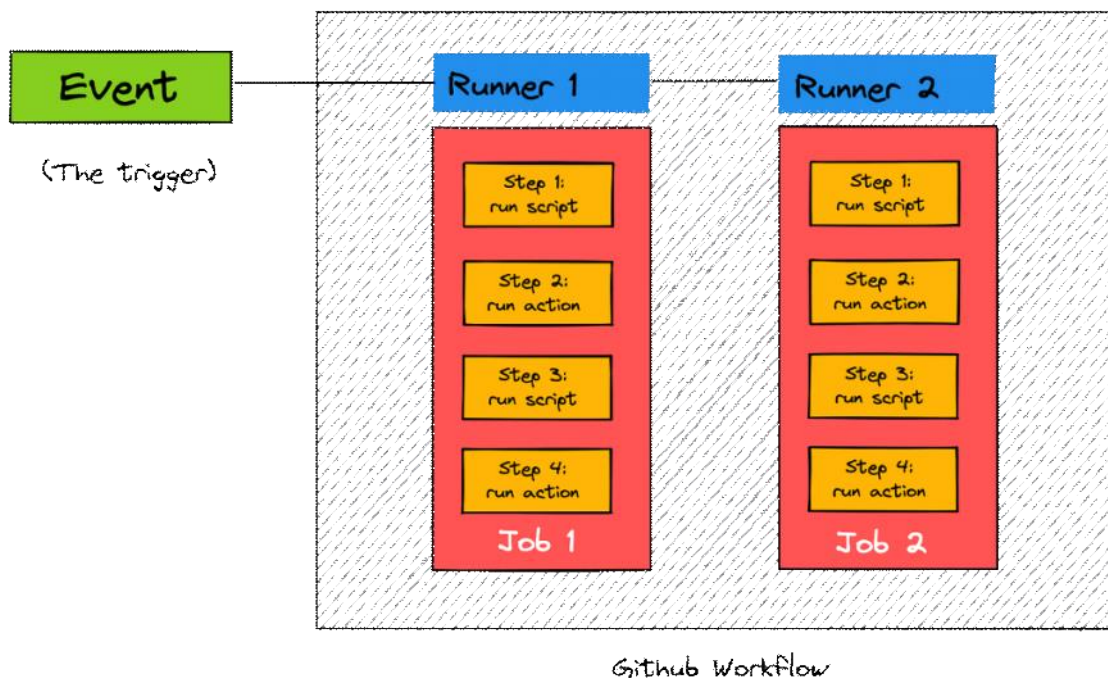
Wednesday, November 19, 2025

11:34 AM

Overview

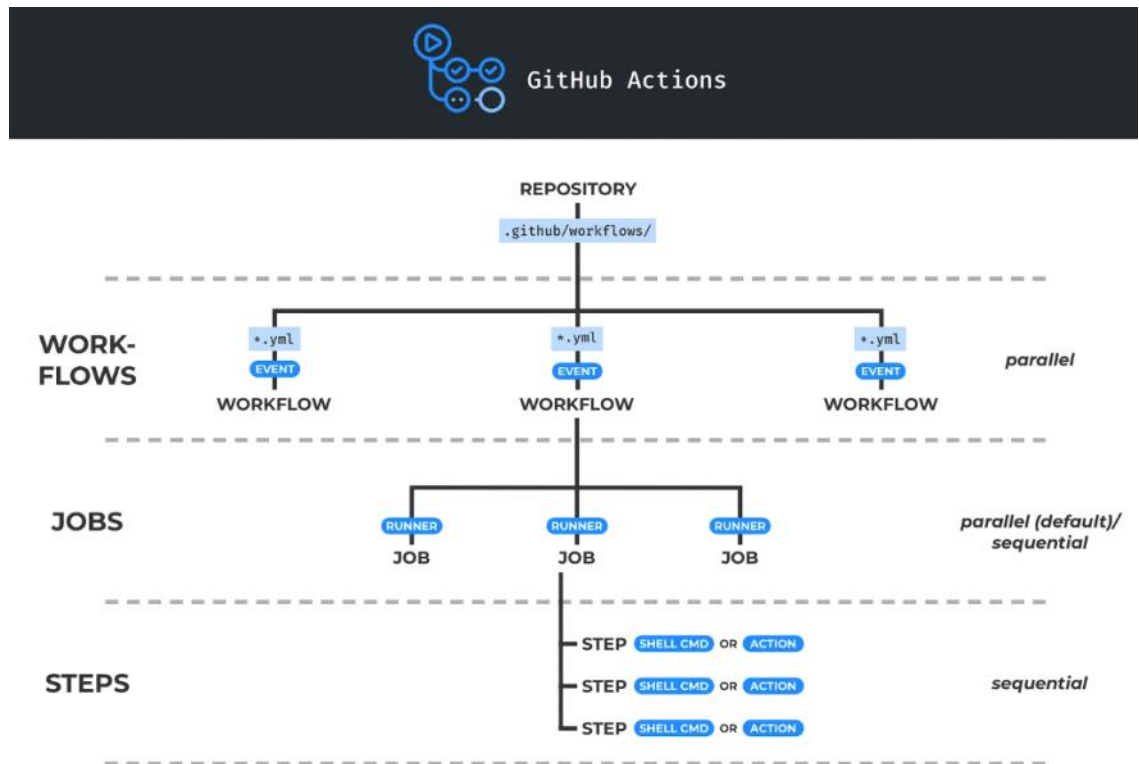
1. Helps to automate software **development workflows** right in your repository.
2. **Actions** perform **any job** you'd like:
 - a. **CI/CD**
 - i. Workflows that **build** and **test** every pull request to your repository, or **deploy merged pull requests** to production.
 - b. Contributor joined repo
 - c. Code reviews
 - d. Branch management
 - e. Issue related events
 - i. Assigning labels
 - ii. Auto reply etc.
 - f. You can create a **workflow** for almost any type of **event**
 - i. Pull request
 - ii. Push
 - iii. Open an issue

Components of GitHub Actions



Workflows are configurable automated process that will run one more jobs. These workflows are triggered:

- a. by an **event in your repository**
- b. **Manually**
- c. At a defined **schedule**



What to Achieve

GitHub Flow with GitHub Actions



Documentation:

Workflow syntax

Pre-build actions

Contexts

Introduction YAML

Wednesday, November 19, 2025 11:37 AM

Comparison between different data serializations

XML	JSON	YAML
<pre><Servers> <Server> <name>Server1</name> <owner>John</owner> <created>123456</created> <status>active</status> </Server> </Servers></pre>	<pre>{ Servers: [{ name: Server1, owner: John, created: 123456, status: active }] }</pre>	<pre>Servers: - name: Server1 owner: John created: 123456 status: active</pre>

Key value pair

YAML is written in key-value pairs. A YAML mapping of a key to a value looks like this

```
Key1: value1
Key2: value2
Key3: value3
...
```

Arrays or Lists

```
# Non indented
names:
- "sajid"
- "kashif"
- "ali"
```

```
# Indented
names:
  - "sajid"
  - "kashif"
  - "ali"
```

Dictionaries

```
faculty:  
  name: sajid  
  email: sajid.ali@giki.edu.pk  
  department: FCSE
```

```
staff:  
  name: sajid  
  email: sajid.ali@giki.edu.pk  
  department: FCSE
```

Dictionaries and Arrays/Lists

```
Employees:  
  - faculty:  
    name: sajid  
    email: sajid.ali@giki.edu.pk  
    department: FCSE  
  
  - staff:  
    name: sajid  
    email: sajid.ali@giki.edu.pk  
    department: FCSE
```

Note: spaces are important in YAML format. Take a moment to understand the examples discussed in a class

For more details, visit [here](#)

Workflow - Example1

Wednesday, November 19, 2025 11:37 AM

```
1  name: First Workflow Using GitHub Actions
2
3  run-name: ${{ github.actor }} has triggered this workflow using ${{ github.event_name }} on ${{github.repository}} repo.
4
5  on:
6    - push
7    - pull_request
8
9  jobs:
10   first-job:
11     runs-on: ubuntu-latest
12     steps:
13       - name: Checking out to the repository
14         uses: actions/checkout@v4
15       - name: Printing the contents of a file
16         run: cat firstfile.txt
17   second-job:
18     runs-on: ubuntu-latest
19     steps:
20       - name: Echo any text using echo bash command
21         run: echo "Hi class, hope you all had a good mid semester break"
```

Please follow the documentation [here](#)

Workflow - Example2

Wednesday, November 19, 2025 11:38 AM

```
1  name: First Workflow Using GitHub Actions
2
3  run-name: >+
4      ${{ github.actor }} has triggered this workflow \
5      using ${{ github.event_name }} event \
6      on branch ${{ github.ref }} of ${{github.repository}} repo.
7
8  on:
9      push:
10         branches:
11             - main
12     pull_request:
13         branches:
14             - main
15
16  jobs:
17     first-job:
18         runs-on: ubuntu-latest
19         steps:
20             - name: Checking out to the repository
21               uses: actions/checkout@v4
22             - name: Printing the contents of a file
23               run: cat firstfile.txt
24     second-job:
25         runs-on: ubuntu-latest
26         steps:
27             - name: Echo any text using echo bash command
28               run: echo "Hi class, hope you all had a good mid semester break"
```

Please follow the documentation [here](#)

Workflow - Example3

Wednesday, November 19, 2025 11:38 AM

```
---  
  
name: "firstworkflow with devops class"  
  
run-name: |  
    "firstworkflow with devops class triggered by ${github.actor}}  
    with ${github.event_name}} event  
    on branch ${github.repository}}"  
  
on:  
push:  
    branches:  
        - main  
pull_request:  
    branches:  
        - main  
workflow_dispatch:  
  
jobs:  
    first-job:  
        runs-on: ubuntu-latest  
        steps:  
            - name: Checking out to a repository  
              uses: actions/checkout@v4  
            - name: Printing the content of the file  
              run: cat firstfile.txt  
            - name: Executing script  
              run: bash a-simple-script.sh  
    second-job:  
        if: ${github.event_name == 'pull_request' }}  
        runs-on: ubuntu-latest  
        steps:
```

```
- name: Installing php7
  uses: shivammathur/setup-php@v2
  with:
    php-version: "7.3"
- name: Installing nodejs
  uses: actions/setup-node@v4
  with:
    node-version: "17"
- name: Checking the versions of php and nodejs
  run: |
    php --version
    node --version
```

...

CICD For React App

Wednesday, December 10, 2025 9:55 AM

<https://github.com/msajidaligik/fall2025-cs423-cicd-node-app.git>



fall2025-c
s423-cicd-...

Demos

Wednesday, December 10, 2025 9:57 AM



workflows

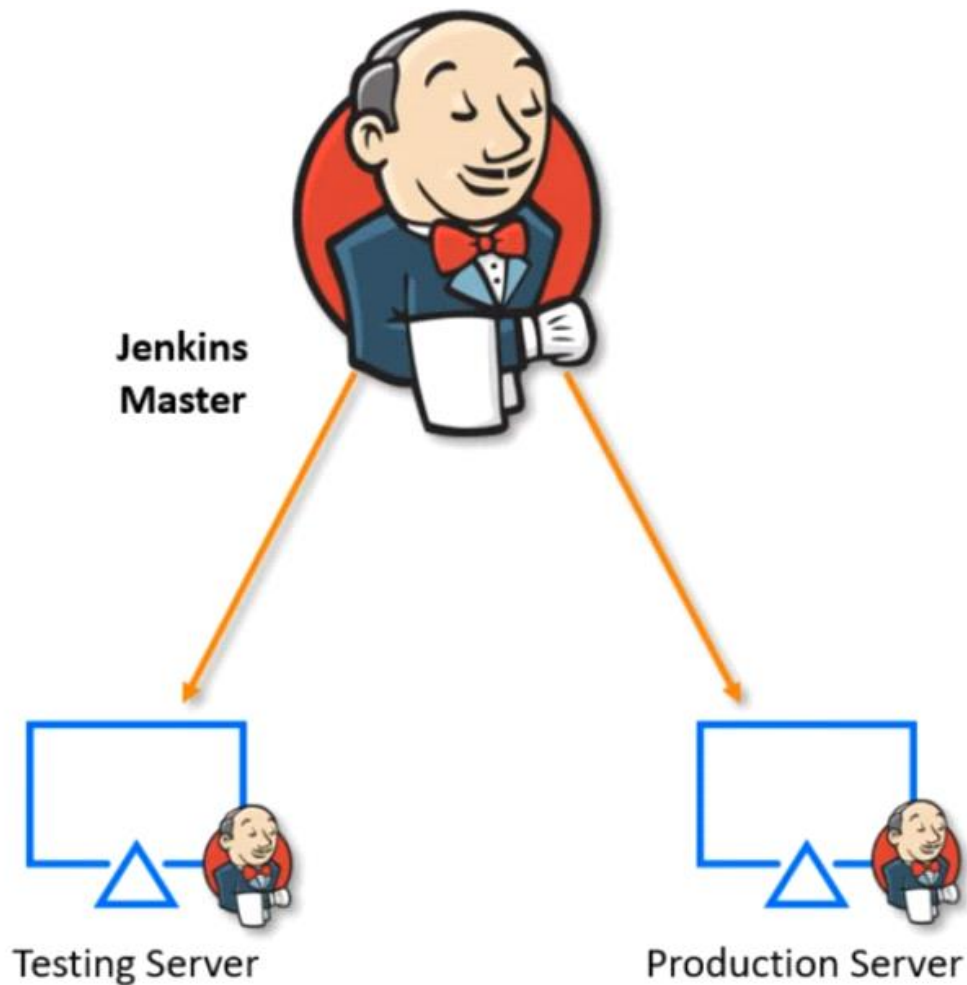


workflows

Jenkins – CI/CD Tool

Wednesday, November 19, 2025 11:40 AM

Jenkins Master/Slave Architecture



Jenkins Installation

Please follow the detailed steps defined here in order to install jenkins

<https://www.jenkins.io/doc/book/installing/linux/>

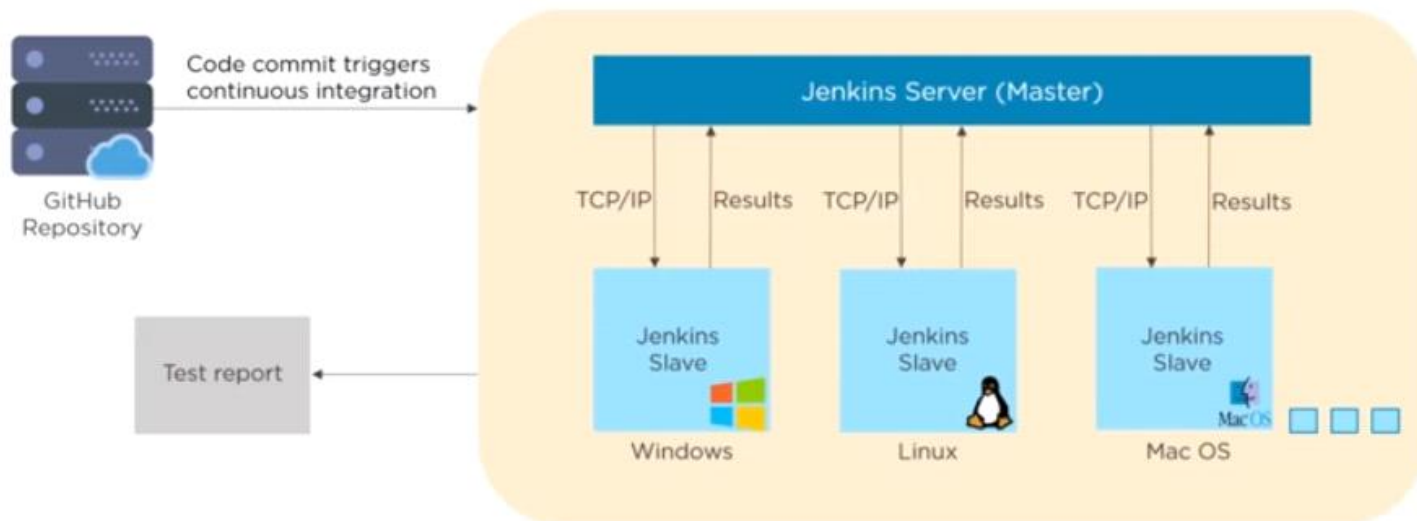
Manage Nodes and Clouds on Jenkins

1. Go to dashboard
2. Click --> Manage Jenkins
3. Click --> Manage Nodes and Clouds
4. Click --> New Node
5. Follow up the form

To add known hosts

1. `ssh-keyscan -H <hostname or IP> >> var/lib/jenkins/.ssh/known_hosts` file

CI/CD Build



Installation

Wednesday, November 19, 2025 11:41 AM

Please follow the detailed steps defined here in order to install jenkins

<https://www.jenkins.io/doc/book/installing/linux/>

Tentative Tasks

Wednesday, November 19, 2025

11:42 AM

1. Managing Jenkins
2. Configurations
3. Credential Manager
4. Creating jobs and pipelines
5. Managing nodes
6. Managing plugins
7. Running jobs on certain nodes
8. Build tasks
9. Post build tasks
10. CI/CD Pipeline

Scripted Pipeline

Wednesday, November 19, 2025 11:42 AM

Scripted Pipelines

```
node {  
    stage('Getting-Started') {  
        echo 'AoA, this is my first scripted pipeline'  
    }  
}
```

Declarative Pipelines --> Let's explore in a demo